

Technology Stack

Date	6 June 2026
Team ID	XXXXXX XXXXXX
Project Name	Rising Water
Maximum Marks	2 Marks

Define Your Technology Stack:

Identify the programming languages, frameworks, databases, front-end tools, back-end tools, and APIs that you will use to build your solution. A well-chosen technology stack ensures that your application is scalable, maintainable, and aligned with your project requirements.

Fill out the table below to detail the specific technologies chosen for each component of your architecture and provide a brief justification for your choice.

Technology Stack Details

S.No	Architecture Component / Layer	Technology Chosen	Justification / Purpose
1	Frontend / Client-Side <i>(e.g., React, Angular, HTML/CSS)</i>	<i>HTML5, CSS3, Bootstrap, JavaScript</i>	<i>Responsive and easy-to-use user interface</i>

2	Backend / Server-Side <i>(e.g., Node.js, Python/Django, Java)</i>	<i>Python, Flask</i>	<i>Lightweightframework suitable for ML deployment</i>
3	Machine Learning	<i>Scikit-learn (Random Forest)</i>	<i>Accurate flood prediction model</i>
4	Cloud / Hosting / Deployment <i>(e.g., AWS, Azure, Heroku, Docker)</i>	<i>Render / Streamlit Community Cloud</i>	<i>Free and easy deployment</i>
5	Version Control & CI/CD	<i>Git & GitHub</i>	<i>Code management and collaboration</i>